1. Technical Data Sheet

HEAT SHIELD

Silicone Coated Fiberglass Fabric

SILICONE COATED FIBERGLASS FABRIC (ONE SIDE COATED)

ITEMS	DATA		
MODEL CODE	SG3784A180GY		
WEAVE	SATIN		
WIDTH	1000 TO 1800mm		40 to 72 inch
TOTAL THICKNESS	0.85 ± 0.09mm		0.033 ± 0.0035 inch
COAT STYLE	**ONE SIDE** GREY SILICONE COATING		
AREA WEIGHT	**990 ± 20 g/m²**		29.20 ± 0.59 OZ/YD2
TENSILE STRENGTH	WARP	≥ 2600 N/25mm	≥ 594 lbs/inch
	WEFT	≥ 2900 N/25mm	≥ 663 lbs/inch
MAX. WORKING TEMPERATURE	BASIC FABRIC : 550°C SILICONE COATING : 260°C		

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HEAT SHIELD

Aluminised Textured FORTAGLAS ™ "E" Glass Fabric

Product Identification Reference	**GKAA2006**
Description	Aluminised Textured FORTAGLAS ™ "E" Glass fabric.
Weave	3 x 3 stepped twill
Finish	Aluminium foil laminated to both sides
Weight (g/m²)	2200 nominal
Thickness (mm)	2.9 nominal
Cloth construction Ends / dm Picks / dm	 66 nominal 70 nominal
Tensile strength Warp (N / 50mm) Weft (N / 50mm)	 4700 nominal 5000 nominal
Flammability (EN532 indicative)	Classed as Non-flammable
Roll length (metre)	25 nominal
Roll width (cm)	100 nominal

The information contained in this data sheet is believed to be true at the time of printing. Any statements contained or inferred to within are an expression of opinion and presented without guarantee. It is up to the user to determine suitability of use, or potential patent infringement for specific applications.

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HEAT SHIELD

Ceramic Fiber Blanket

Description:

CBLK96/CBLK128 is a needled blanket made from our ceramic bulk fiber which can withstand temperature up to 1400C degree (depends on different quality grades). The blanket is lightweight, flexible, and available in a wide variety of thicknesses, widths and densities.

Application:

-Insulation and linings for furnaces, kilns, generators, reformers, boilers, etc.
-High-temperature pipe insulation, casting mold insulation, etc.
-High-temperature seals and gaskets, furnace door seals, expansion joint seals and filtration

Technical Data

Item		COM	STD	HP	HA	HZ
Specification temp (ºC)		1260	1260	1260	1400	1400
Working temp (ºC)		＜1000	1000	1100	1200	1350
Color		White	White	White	White	White
Dens (kg/m3)		96 128	96 128	96 128	96 128	96 128
Rate of liner (%) (24h. 128kg/m3)		-4 (1000ºC)	-3 (1000º)	-3 (1100ºC)	-3 (1200°C)	-3 (1350ºC)
Rate of thermal Conductivity (W/m.K) (128kg/m3)		0.09 (400ºC) 0.16 (800ºC)	0.09 (400ºC) 0.16 (800ºC) 0.20 (1000ºC)	0.09 (400ºC) 0.16 (800ºC) 0.20 (1000ºC)	0.12 (600ºC) 0.20 (1000ºC)	0.15 (800ºC) 0.20 (1000ºC)
Rate of thermal Conductivity (W/m.K) (96kg/m3)			0.138 (400ºC) 0.245 (800ºC) 0.30 (1000ºC)			
Standard roll dimension (mm)		12.5 x 610 x 14400, 25 x 610 x 7620, 50 x 610 x 3810				
Tensile strength (Mpa) (128kg/m3)		0.04	0.05	0.04	0.04	0.04
Tensile strength (Mpa) (96kg/m3)			0.04			
Chemical Composition (%)	Al2O3	44	46	47-49	52-55	39-40
	Al2O3+SiO2	96	97	96	99	-
	Al2O3+SiO2+ZrO2	-	-	-		99
	ZrO2	-	-	-		15-17
	Fe2O3	＜1.2	＜1.0	0.20	0.20	0.20
	Na2O+K2O	＜＝0.5	＜＝0.5	0.20	0.20	0.20

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Velcro type Fasteners (Hook)

Construction characteristiques	
Basic construction	Woven
Base material	Nylon
Standard binder backing	Synthetic resin
Width:	
< 50 mm	Nominal +/- 1 mm
>= 50 mm	Nominal +/- 1,5 mm
>= 100 mm	Nominal +/- 2 mm
Hook / cm^2	65 +/- 5
Overall thickness	2.25 +/- 0.25 mm
Weight / m^2	400 g/ m^2 +/- 15%
Usage characteristiques	
Peel Strength with Loop std*	Average 2.0 N/cm Minimum 1.3 N/cm
Shear Strength with Loop std*	Average 10.3 N/cm^2 Minimum 7.3 N/cm^2
Breaking Strength *	Minimum 260 N/cm
Shrinkage after washing (3 x 60°C)	Maximum 4%
Max. Service Temp.	177°C
Melting point	250 °C

The information included in this Technical Sheet is based on reliable tests and trials. The average values correspond to the averages to the previous year to the edition of this document. Given the diversity of uses of our products we advice our customer to assure themselves that the product meets requirements of their application. The responsibility for the application and use of the product remains with the customer.

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Doc # HS-0231-17 dated 18.05.17

HEAT SHIELD

Velcro type Fasteners (Loop)

Construction characteristiques	
Basic construction	Woven
Base material	Nylon
Standard binder backing	Synthetic resin
Width:	
< 50 mm	Nominal +/- 1 mm
>= 50 mm	Nominal +/- 1,5 mm
>= 100 mm	Nominal +/- 2 mm
Overall thickness	1.5 +/- 0.25 mm
Weight / m^2	300 g/ m^2 +/- 15%
Usage characteristiques	
Peel Strength with Loop std*	Average 2.0 N/cm Minimum 1.3 N/cm
Shear Strength with Loop std*	Average 10.3 N/cm^2 Minimum 7.3 N/cm^2
Breaking Strength *	Minimum 210 N/cm
Shrinkage after washing (3 x 60°C)	Maximum 4%
Max. Service Temp.	177°C
Melting point	250 °C

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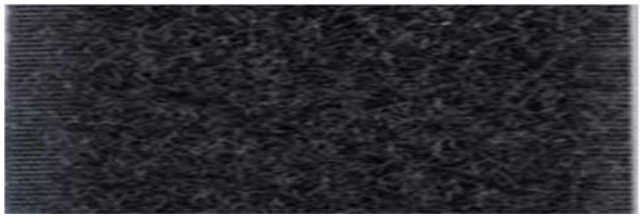

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Doc # HS-0231-17 dated 18.05.17

Stainless Steel Cord End Clip

Description: Cord End Clips

Colour: Clean (Unplated)

Sizes (mm):

Overall Height:	7.20mm
Internal Height:	2.30mm
Overall Length:	10.00mm
Tooth Height (8 in total):	1.50mm
Internal Tooth Height (4 in total):	1.10mm
Internal Diameter at Cord Entry:	6.00mm
Folding Angle:	102º from base
Base Piercing:	None
Side Piercing:	None
Metal Thickness:	0.3mm

Tolerances +/-0.20mm

Weight: 0.446g each

Packing: Cardboard Box/Plastic Bag

Packing Unit: 1,000

Composition: Stainless Steel Grade 304

Clean Finish

The clean finish is a process whereby the surface of the clip is degreased and polished. No residue is left on the surface of the eyelet.

Handling

These parts are made from stainless steel. This is an inert metal and requires no special handling. However, it is advisable to keep the components in a dry environment.

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HEAT SHIELD

Stainless Steel Eyelet & Ring

32 Sail Eyelet 1A/32

Dimensions

Minimum Internal Diameter when Closed	:	19.05mm
Overall Length	:	10.70mm
Flange Thickness	:	1.57mm
Flange Diameter	:	34.34mm
Metal Thickness	:	0.51mm
Weight	:	4.512g each

32 T/O Ring 32/1D

Dimensions

Outside Diameter	:	35.27mm
Hole Diameter	:	21.31mm
Thickness	:	3.06mm
Weight	:	2.47g each

Tolerances	:	+/-0.20mm
Packing	:	Cardboard Box
Packing Unit	:	1,000
Composition	:	SS-304
Finish	:	Clean & Polish

Clean & Polish Finish
Clean and Polish finish is a process whereby the surface of the brass is cleaned of grease and other manufacturing residues. It is then polished to a shine leaving no surface coating.

Handling
These parts are made from brass. This is an inert metal and requires no special handling. However, it is advisable to keep the components in a dry environment in order to prevent tarnishing.

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HEAT SHIELD

Braided Fiberglass Rope

Description:
Braided fiberglass rope is fabricated with high quality texturized fiberglass yarns that will not burn and will withstand continuous exposure to temperatures of 1000°F / 520°C. This material resists most acids and alkalis and is unaffected by most bleaches and solvents. It is highly flexible and conformable.

Specification:

FG102T	Round braided: Dia. 4mm up to 50mm;
FG103T	Square braided: Dia.4x4mm up to 50x50mm;
FG103TREC	Rectangular braided: as required

Application:
It can be used as a gasket or seal on boiler, coke oven, industrial oven and wood stove/pellet stove doors. It is also used for crucible packing and pollution control equipment, pump shaft packing.

Physical Properties:
Yarn filament: 11um
High temp. Resistance: up to 500c degree
Low thermal conductivity: ≤0.035keal/m.h. °C

Chemical Properties:

Item	Unit	Data
SiO2	%	54.1%+/-0.5
Al2O3	%	14.6%+/-0.4
CaO	%	16.6%+/-0.3
MgO	%	4.6%+/-0.3
R2O	%	<0.8%
Fe2O3	%	<0.4%
B2O3	%	8.8%+/-0.5

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HEAT SHIELD

PTFE Sewing Thread

ITEMS	DATA
THREAD DIAMETER	0.45 - 0.50 MM
TEX	14 TEX * 12 PLY
COATING MATERIAL	PTFE
TWIST DIRECTION	Z
TURN PER METER	≥ 280
TENSILE STRENGTH (N)	90 - 110
FILAMENT DIAMETER (UM)	5.5 - 6.5
TEMPERATURE RESISTANCE	550°C
WEIGHT PER ROLL	0.8 - 1.8 KG
COLOR	BROWN
PROPERTIES	It is made of multi-ply fiberglass roving yarn treated with PTFE for smooth and easy processing after high twist compound. It features high tensile strength, good sewing property and high temperature resistance. It widely used in sewing high temperature weave.
APPLICATION	It is mainly used for sealing and insulating products sewing safety clothing, gloves, welding and fire blankets

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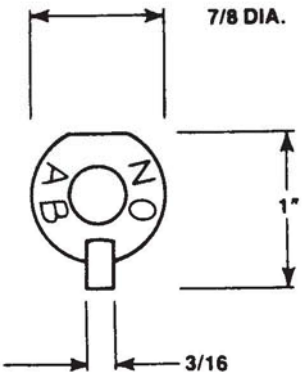
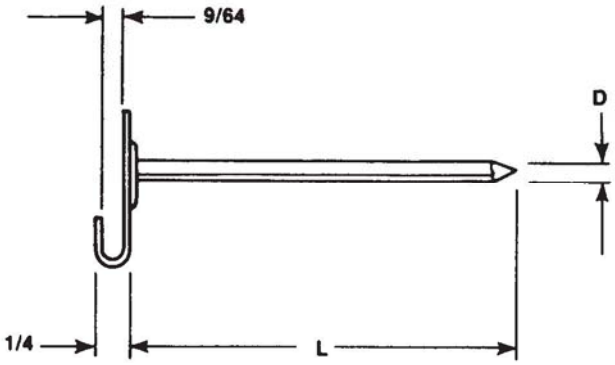
Lacing Anchor





SIZE : D 12 GA & 14 GA are standard. Other sizes are available by special order.

LENGTH : L 2-1/2" & 4-1/2" are standard. Other lengths available by special order.

MATERIAL : Stainless Steel

WASHERS : Self-locking washers are available in a variety of sizes, shapes and materials.

	Washer	Dome Cap
Material	316 stainless steel	Washer : stainless steel CAP-SS : stainless steel dome
Diameter	External dia. : 30mm~38mm For pins dia. 2-2.2-2.7-3-4mm	External dia. : 22mm CAP-2 for pins dia. 2mm CAP-27 for pins dia. 2,7mm

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SS 316 /316L Wire

Stainless Steel Tie Wire is drawn to precision diameter, and then soft annealed for quick field insulation tie-ups.

Stainless Steel Tie Wire comes in different Alloy i.e. T304, T304L, T316, T316L. SS Wire available in size of 0.5mm, 1.0mm, 1.2mm, 1.3mm, 1.5mm, 1.6mm, 1.8mm

SS 316/ 316L Wire				
Diameter	0.5mm, 1.0mm, 1.2mm, 1.3mm, 1.5mm, 1.6mm,1.8mm,			
Chemical Composition	**SS 316**		**SS 316 L**	
Component	**Max (%)**	**Min (%)**	**Max (%)**	**Min (%)**
C	0.07		0.03	
Si	1.00		0.75	
Mn	2.00		2.00	
S	0.015		0.03	
P	0.045		0.045	
Ni	13.00	10.00	14.00	10.00
Cr	18.50	16.50	18.00	16.00
Mo	2.50	2.00	3.00	2.00
N	0.11		0.10	

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2. Material Safety Data Sheet

Material safety data sheet

Section 1-Product and Company Identification

Reference No:	SG3784A180GY
Product description:	Silicone coated Fiberglass Cloth
Revision Date:	01/01/2023
Supplier Details:	Heat Shield P. O. Box 49374, Dubai - UAE

Section 2-Hazards Identification

Classification of the Substance or Mixture

GHS Classification in Accordance with 29 CFR 1910(OSHA HCS):

Health, Skin corrosion/irritation, 3

Health, Specific target organ toxicity - Single exposure, 3

GHS Label Elements, Including Precautionary Statements

GHS Signal Word: WARNING

GHS Hazard Pictograms:

HEAT SHIELD

GHS Hazard Statements:

H316 - Causes mild skin irritation

H335 - May cause respiratory irritation

GHS Precautionary Statements:

P264 - Wash _ thoroughly after handling.

Hazards not Otherwise Classified (HNOC) or not covered by GHS

Route of Entry:	This material may enter the body through inhalation of nuisance dust.
Target Organs:	Respiratory system
Inhalation:	Sore, raspy throat
Skin Contact:	Redness and possible rash; itching
Eye Contact:	Itching and redness

Section 3-Composition/Information on Ingredients

Ingredients:

Cas#	%	Chemical Name
65997-17-3		Fibrous Glass
	0	Compounded silicone coating

Fibrous Glass (CAS 65997-17-3)

Exposure Limits: OSHA PEL 15 mg/m3 (total), 5 mg/m3 (respirable). ACGIH TLV 1 f/cc

Section 4-First Aid Measures

Inhalation: Remove person to fresh air. Seek medical attention if condition persists.

Skin Contact: Rinse with copious quantities of cool water. If rash or itching persists, seek medical attention.

Eye Contact: Rinse with water. Do not rub eyes. Seek medical attention.

Ingestion: N/A

Section 5-Fire Fighting Measures

Flash Point(Method Used): >250℃ by TOC

Flammable Limits

LEL: N/A

UEL: N/A

Extinguishing Media Water, carbon dioxide, dry chemical

Special Fire Fighting Instructions Thermal decomposition of fiber coating may produce an irritating mixture of smoke and fumes.

Unusual Fire and Explosion Hazards None

Section 6-Accidental Release Measures

Material is a solid in roll form. If accidently released, rewind material back onto roll.

Section 7-Handling and Storage

Handling Precautions: Use adequate material handling equipment

Storage Requirements: Store in dry place. Use may be at temperature extremes based on product data, but storage should be at ambient temperature.

Section 8-Exposure Controls/Personal Protection

Engineering Controls: Local exhaust; dust collection

Personal Protective: HMIS PP, B | Safety Glasses, Gloves

Equipment: Safety glasses; cotton gloves; long sleeve clothing

Wash thoroughly with soap and water after handling

Section 9-Physical and Chemical Properties

Appearance:	Grey colored coated fabric	**Odor:**	No Odor
Physical State:	Solid	**Solubility:**	Negligible
Spec Gray/Density:	2.4	**Freezing/Melting Pt:**	＞1000 F
Boiling Point Vapor:	N/A	**Vapor Density:**	N/A
Vapor Pressure:	N/A	**pH:**	N/A

Section 10-STABILITY AND REACTIVITY

Chemical Stability:	Material is stable.
Conditions to Avoid:	None known
Materials to Avoid:	Strong oxidizing agents.
Hazardous Decomposition:	Carbon monoxide; carbon dioxide
Hazardous Polymerization:	Will Not Occur.

Section 11-Toxicological Information

Direct contact with fiberglass materials or exposure to airborne fiberglass dust may irritate the skin, eyes, nose and throat.

Fiberglass can cause itching due to mechanical irritation from the fibers. This is not an allergic reaction to the material. Breathing fibers may irritate the airways resulting in coughing and a scratchy throat. Some people are sensitive to the fibers, while others are not.

Section 12-Ecological Information

No known hazards except for airborne fibers caused by nuisance dust. 10 milligrams per cubic meter for fiber diameters less than 7 microns.

Section 13-Disposal Considerations

Incineration preferred in a federal, state, or local approved facility.

Section 14-Transport Information

None special required.

Section 15-Regulatory Information

Component (CAS#) [%] - CODES

Fibrous Glass (65997-17-3) [n/a%] TSCA

Compounded silicone Coating (0) [n/a%]

Regulatory CODE Descriptions

TSCA = Toxic Substances Control Act

Section 16-Other Information

NFPA: Health = 1, Fire = 0, Reactivity = 0, Specific Hazard = n/a

HMIS Ⅲ: Health = 1, Fire = 0, Physical Hazard = 0

HMIS PPE: B-Safety Glasses, Gloves

Disclaimer:

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HEAT SHIELD

Material Safety Data Sheet

Revision Date: June 19, 2020

1. PRODUCT

PRODUCT NAME : CERAMIC FIBER BLANKET/ CERAMIC FIBER TAPE

PRODUCT DESCRIPTION : Ceramic fiber, Aluminosilicate Refractory Fiber, Refractory Ceramic Fiver (RCF)

2. INFORMATION ON INGREDIENTS

CERAMIC FIBER AND OTHER SIMILAR WOOLS BEING CERAMIC FIBRE BLANKET/ CERAMIC FIBER TAPE

3. HAZARDS IDENTIFICATION

Warning dust from this product generated by handing may cause skin, eye and respiratory tract irritation. Possible hazards depend on duration and level of exposure.

Hazard ratings :

Hazardous Materials information system (HMIS) Ratings:
Health: 1 Flammability: 0 Reactivity: 0 Personal Protection Index: X

Possible effects on health: prolonged and repeated inhalation of aluminosilicate dust may cause chronic effects on respiratory system such as bronchitis, asthma, and emphysema signs

And symptoms of excessive exposure:

Eye contact: physical irritation
Skin contact: physical irritation
Ingestion: temporary irritation to gastrointestinal tract
Inhalation: pulmonary dysfunction

4. FIRST AID MEASURES

EYE CONTACT :
If eyes become irritated, wash immediately with large amounts of lukewarm water for at least 15 minutes. Eyelids should be held away from the eyeball to ensure thorough rinsing. Do not rub eyes. Get medical attention if irritation persists.

SKIN CONTACT :
If skin becomes irritated, do not rub or scratch exposed skin. Wash area of contact thoroughly with soap and water. Using a skin cream or lotion after washing may be helpful. Change into clean clothing.

INGESTION : relocate affected individual to an environment of clean and fresh air. Drink plenty of water seek medical help if symptoms persist.

INHALATION : remove affected individual to a dust free place, seek medical help if irritation persists.
Notes to physicians: Skin and respiratory effects are the result of mechanical irritation: fiber exposure does not result in allergic manifestations.

5. FIRE FIGHTING MEASURES

Non-combustible(does not burn) product.
Auto-ignition temperature: none
NFPA Unusual Hazards: none
Unusual fire and explosion hazards: none
Extinguishing Media: use proper extinguishing media for the surrounding fire.
Fire fighting protective equipment: wear full bunker gear including positive pressure self-contained breathing apparatus.

6. ACCIDENTAL RELEASE MEASURES

Avoid creating airborne dust. Maintain routine housecleaning procedures. Vacuum only with HEPA filtered equipment, if sweeping is necessary, use a dust suppressant and keep material in closed containers. Do not use compressed air for clean-up. Workers should wear gloves, goggles and approved respirator. Avoid clean-up procedures that could cause water pollution.

7.HANDLING AND STORAGE

CLEAN-UP :
Clean up dust carefully. Use wet sweeping or high efficiency vacuum to remove dust. Do not use compressed air.
During after-service removal activities, wet exposed material frequently to minimize airborne dust. A surfactant may be added to the water to improve the wetting process. Use only enough water to wet the insulation. Do not allow water to accumulate on floors.

EMPTY CONTAINERS :
Product packaging may contain residue. Do not reuse.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Components OSHA supplier
Alumino--silicate fiber None established

ACGIH TLV: Aluminosilicate fiber—none established

For Alumino--silicate fiber, refer to OSHA guidance regarding "Particulates not otherwise Regulated"(PNOR). Control airborne dust levels as follows:

Components Particle size OSHA

PNOR total dust 15mg/m3

Respirable dust 5mg/m3

ACGIH particulates not otherwise classified (PNOC)—INHALABLE PARTICULATE:10mg/m3. RESPIRABLE PARTICULATE: 3mg/m3

Other Occupational Exposure Levels (OEL)

RCF-related occupational exposure limits vary from country to country. Listed here are a few regulatory OEL examples: Australia--0.5f/cc; Austria—0.5f/cc; Canada—0.5 to 1 f/cc; Denmark—1 f/cc

France—0.6 f/cc; Germany—0.5 f/cc; Netherlands—1 f/cc; United Kingdom—2 f/cc

Example is: RCFC REG 0.5 f/cc. The objectives and criteria underlying each of these OEL decisions also vary. The evaluation of occupational exposure limits and determining their relative applicability to the workplace is performed on a case-by-case, by a qualified industrial hygienist.

Engineering Controls :

Use engineering controls such as ventilation and dust collection devices to limit airborne fiber concentrations to the minimum attainable level.

Eye Protection:

In case of overhead work, wear goggles or safety glasses with side shields to prevent eye contact.

Skin Protection:

Wear gloves, head covering and full body clothing as necessary to prevent skin irritation.

Respiratory Protection:

When effective engineering and administrative controls are insufficient, the use of appropriate respiratory protection, pursuant to the requirements of OSHA 1910.134 and 29 DFR 1926.103 for the particular hazard or airborne concentrations in the work place, it is recommended. For dust concentrations below applicable exposure limit value.

9. PHYSICAL AND CHEMICAL PROPERTIES

Oxidizing properties None	Vapor density: not applicable
Odor None	Water solubility: not applicable
Chemical family: Aluminosilicate fibers	PH: not applicable
Chemical Indexes: $Al_2O_3+SiO_2$ >97%	Boiling Point: not applicable
Al_2O_3 >46%	Melting point: 3200℉
Fe_2O_3 <1.0%	
Na_2O+K_2O<=0.5	

10. STABILITY AND REACTIVITY

Chemical stability: stable under conditions of normal use.
Incompatibility: hydrofluoric acid, and concentrated alkali.

11. TOXICOLOGICAL INFORMATION

Epidemiological studies conducted by institution of human environment protection in China has provided no evidence that there is a direct cause-and effect relationship between cumulative exposure to aluminosilicate fibers and lung cancers or particular pulmonary diseases.
However recent toxicological experiments using physiological exposure method(inhalation)have produced findings of respiratory disease in rodents, Aluminosilicate refractory fiber has found to be a rodent carcinogen under the conditions that the rodents are exposed to high levels of the material(75 - 115fibers/cc)on a basis of lifetime duration.

12. ECOLOGICAL INFORMATION

No data is available on adverse effects of the material on the environment.

13. DISPOSAL CONSIDERATIONS

As produced, this material is usually accepted for disposal at most sites licensed for the disposal of industrial waste. Check applicable regulations and waste site policies prior to disposal. Waste should be paced containers for disposal.
In case of contamination, by other materials classified as hazardous waste, expert guidance should be sought.

14. TRANSPORT INFORMATION

Product should remain in sealed containers during transportation.

15. REGULATORY INFORMATION

CERCLA: the aluminosilicate fibers of this product have an average diameter of 2-4 um and are not considered CERCLA hazardous substances (CERCLA 40 CFR302)
Clean Air Act (CAA): thermal insulation fibers are composed of (RCF) with an average diameter greater than 1 micron, and therefore are not considered hazardous air pollutants.
Toxic Substances Control Act (TSCA): all substances in this product are listed, as required, on the TSCA chemical inventory.
State Regulations
California: aluminosilicate fiber has not been listed by the State of California on proposition 65, the Safe Drinking Water and Toxic Enforcement Act of 1986.
New Jersey: Chemicals which are listed as special health hazard substances as defined in New Jersey worker and Community Right to know Act, New Jersey Administrative code, title 8, Department of health, Chapter 59, Subchapter 10

Pennsylvania: aluminosilicate fiber is not listed as a special health hazard substance as defined in Pennsylvania Right-to-Know Law, Section 3800.

Chemical Name	CAS Number
None	142844-00-6

International Regulations:
Canadian Workplace Hazardous Materials Information System(WHMIS):
No Canadian Workplace Hazardous Materials Information System categories apply to this product.

Canadian Environmental Protection Act(CEPA):
All substances in this product are listed, as required, on the Domestic Substances List(DSL). Chemicals which are listed on the Non-Domestic Substances list:

Chemical Name	CAS Number
None	

16. OTHER INFORMATION

Removal after service: Under sustained and steady high temperature over 1800°F, this material will possibly transform to crystalline silica(ciystobalite) in exposed portions. Prolonged or repeated exposure to respirable crystalline silica dust may lead to lung diseases. IARC has listed crystalline silica in Category 2A a probable carcinogen (" crystalline silica inhaled in the form of quartz or cristobalite from occupational source is carcinogenic to humans" IARC monograph 68, June 1997 p 210—211). The permissible exposure limit(PEL) set by OSHA for respirable cristobalite is 0.05mg/m3. whenever possible follow section 8 procedures for exposure controls and personal protection.

Abbreviations:
CERCLA: comprehensive environmental response compensation and liability act of 1980
CAS; Chemical abstracts service
f/cc: fibers per cubic centimeter
HMIS: Hazardous Material information system
Mg/m3: Milligrams per cubic meter of air
NIOSH: National institute for Occupational Safety and Health
OSHA: Occupational Safety and Health Administration
PEL: permissible exposure limit
SARA: super amendments and reauthorization act
TSCA: toxic Substances Controls Act

HEAT SHIELD

Material Safety Data Sheet

1.DESCRIPTION: Velcro / Hook&Loop

2.INGREDIENT:

Ingredient/Compose	Content	Component name	Note
Hook	100% Nylon	Nylon 6	Ground: Nylon6
Look		Nylon 6	

3.SPECIFICATION:

Item	Unit	Factory Standard			BS7271 Standard
		0 Time	5000 Times	10000 Times	
Release Force	g /cm	255	210	170	135 < 0 Time
Pullout Strength	g/ cm^2	865	655	525	850 < 0 Time

Colour Fastness(Do not contain fluorescent)	Actuality	Result	Test Method
Colour fastness to light	Shade Change	Above 4 Level	CNS-1493 or JIS-L-0842
Colour fastness to washing	Shade Change	Above 4 Level	CNS-1494 or JIS-L-0844 No.:A-4
	Contaminate	Above 4 Level	
Colour fastness to Hot water washing	Shade Change	Above 4 Level	CNS-1495 or JIS-L-0845 No.:3
	Contaminate	Above 4 Level	
Colour fastness to Sweat	Shade Change	Above 5 Level	CNS-1496 or JIS-L-0848 No.:C
	Contaminate	Above 4 Level	
Fastness to rubbing	Shade Change	Above 4 Level	CNS-1499 or JIS-L-0849 No.:2
	Contaminate	Above 4 Level	

4.HAXARDS SUMMARIZING:

Invasion Route	---
Health Risk	---
Environment Hazards	---
Explosive Danger	Flammable, keep away from fire or flame.

5.EMERGENCY TREATMENT:

Skin Exposure	---
Eyes Exposure	Mention eyelid immediately,and washing with flowing water.
Inhalation/Ingestion	---
Severe Symptoms	In case of fire and after burning, irritation will be generated and hurt skin.
Hazard Warning	Be irritating to the respiratory tract and skin,even life-threatening,should seek medical care immediately
Hazardous Characteristics	Be flammable in case of open flame and high temperature .

6.FIRE PROTECTION:

Suitable fire extinguisher	Dry chemical powder/Foam/Water spray mist
Special Extinguish Procedure	Water spray mist cooling,container or building that near the fire place and exposed to fire.
Protective Device	Must wear air breathing apparatus,Plus a flash fire resistant aluminum cladding coat if necessary.

Other protective	Forbid to smoke in the workplace,should ventilate.
Extinguishing method	1)Firemen must be wearing a gas mask and wear the uniform; 2)Spray at the burning area,Spray cooling until the fire extinguishing; 3)If the container make sound in fire,should move out immediately; 4)Extinguishant/Nepheloidwater/Foam/Dry powder/Carbon dioxide/Sand.

7.ACCIDENTAL RELEASE MEASURES:

Accidental Relaese Measures Note	---

8.PLACE AND STORE:

Place Note	Ventilate,fireproofing,heat source,moistureproof,forbid to smoke in the workplace.
Store Note	Store in a cool ventilated dry place,Fire protection and voltage.Shall not be close to the heat source and acid and alkali corrosive medium,avoid direct sunshine,the store area should be equipped with corresponding varieties and number of fire equipment.

9.PHYSICOCHEMICAL PROPERTY:

Appearance	Solid
Heat Resistance	Melting Point: 230℃ for Hook, 180℃ for Loop; The Hook starts soften at 180 ℃,and melting from 210~220℃.
Cold Tolerance	Can be not workable under -30℃.
Temperature Effect	When the temperature reaches 90%RH,there's a significant decreasethe for tensile force,but will be fully restored to normal after drying.
Main Application	Apparel clothing,Medical supplies,Baby products, Home supplies,Handbag,Toy,Tent,Shoes and Hats,Artware,Stationery,Sports equipment gear,Electronic hardware plastic products
Main Convenient Efficacy	Replace zipper,fastener,pin, retaining ring,Hook.

10.STABILITY AND REACTIVITY:

Stability	It's stable under normal conditions。
Special reaction hazard decomposition	After decomposition or combustion by heat,it has irritant gas and smoke.

11.TOXICOLOGICAL INFORMATION:

Acute/chronic poison	---
Carcinogenicity	---
Stimulating	After decomposition or combustion by heat,it has irritant gas and smoke.

12.WASTE TREATMENT:

Waste Treatment Note	---
Waste disposal methods	Based on relevant rules and regulations.

13.TRANSPORT INFORMATION:

Packing Style	Carton Box
Handling Information	It is strictly prohibited to dleivery with oxidant and any corrosive materials. Should be isolated with power supply and combustion source.

HEAT SHIELD

Material Safety Data Sheet

Section Ⅰ Revision Date: Mar 18, 2017

Product Name : Fiberglass Fabrics

Product Description : Fiberglass fabrics, Texturized fiberglass fabrics, Rope, Tape, Sleeves and other textiles.

Section Ⅱ-Composition/Information on Ingredients

Hazardous Ingredients	Weight %	OSHA-PEL	ACGIH-TLV	Other
Fiberglass, continuous filament	≥ 96.5	*	10mg/m^3 8-hr TWA	3 x 10 fibers/m^3 10-hr TWA(NIOSH)
Non-hazardous Ingredients				
Sizing	≤ 3.5	———— None Established ————		

*OSHA has not established a specific PEL for fibrous glass. It is considered to be a “particulate not otherwise regulated” (PNOR) and is covered under the OSHA nuisance dust PEL’s of 5 mg/m^3 for the respirable dust fraction and 15 mg/m^3 for the total dust fraction for an 8-hr TWA (Time Weighted Average).

Section Ⅲ-Hazardous Identification

Primary Routes of Exposure Inhalation and skin contact

Health Hazards (acute & chronic effects and symptoms of overexposure)

Acute

Inhalation-Inhalation of dusts and fibers may result in irritation of the upper respiratory tract (mouth, nose and throat.)

Skin Contact-Skin contact with fibers and dust may produce temporary mechanical irritations.

Ingestion- Temporary mechanical irritations of the digestive tract. Observe individual. If symptoms develop, consult a physician.

Chronic

See carcinogenicity section below. There is no known health effects associated with chronic exposure to this product.

Carcinogenicity

Hazardous Ingredients	ACGIH	IARC	NTP	OSHA
Fiberglass continuous filament	No	No	No	No

*IARC-In June 1987 the International Agency for research on Cancer (IARC) categorized fibrous continuous filaments as not classifiable with respect to human carcinogenicity(Group 3). The evidence from human as well as animal studies was evaluated by IARC as insufficient to classify fiberglass continuous filaments as a possible, probable, of confirmed cancer causing material.

Medical Conditions Aggravated by Exposure Persons with a history of chronic respiratory or skin conditions that are aggravated by mechanical irritants may be at increased risk for worsening their condition from exposure during use of this product.

Section Ⅳ-First Aid Measures

Inhalation Move individual to fresh air. Seek medical attention if irritation persists.

Skin Contact Wash with mild soap and running water. Use a washcloth to help remove fibers. To avoid further irritation, do not rub or scratch irritated areas. Rubbing or scratching may force fibers into the skin. Seek medical attention if irritation persists.

Eye Contact Flush eyes with flowing water for at less 15 minutes. Seek medical attention if irritation persists.

Ingestion N/A

Section Ⅴ-Fire Aid Measures

Flash Point (° F) N/A

Auto Ignition Temperature(° F) N/A

Flammability Limits (%) LEL: N/A UEL: N/A

Extinguishing Media Water, foam, carbon dioxide, dry chemical

Special Fire Fighting Instructions In sustained fire, self-contained breathing apparatus should be worn.

Unusual Fire and Explosion Hazards None known

Section Ⅵ-Accidental Release Measures

Action To Take For Spills For sold product no applicable. For dusts and fibers generated during fabrication, vacuum up and containerize.

Section Ⅶ-Handling, Storage and Disposal

Ventilation General dilution ventilation and /or local exhaust ventilation should be provided,

as necessary, to maintain exposures below PEL'S or TLV'S. ADEQUATE VENTILATION MUS BE PROVIDED AT ELEVATED TEMPERATURE.

Respiratory Protection A properly fitted NIOSH/MHSA approved dust respirator such as 3M model 8710 or model 9900(In high humidity environment)or equivalent should be used when: high dust levels are encountered; the level of glass fibers in the air exceeds the OSHA permissible exposure limits; or if irritation occurs. Use respiratory protection in accordance with your company's respiratory protection program and OSHA regulations under 29 CFR 1910.134.

Eye Protection Safety glasses, goggles or face shields should be worn whenever fiberglass materials are handled.

Work/Hygienic Practices Handle in accordance with good industrial hygiene and safety practices.

- Avoid unnecessary exposure to dusts and fibers.
- Remove fibers from skin after exposure.
- Be careful not to bur or scratch irritated areas. Rubbing or scratching may force the fibers into the skin. The fibers should be washed off. Use of barrier creams can, in some instances, be helpful.
- Use vacuum equipment to remove fibers and dusts from clothing. COMPRESSED AIR SHOULD NEVER BE USED. Always wash work cloths separately and wipe out the washer/sink in order to prevent loose glass fibers from getting in other clothing.
- Keep work area clean of any dust and fibers. Avoid sweeping or using compressed air as these techniques re-suspend dusts and fibers into air.
- Have access to safety showers and eye wash fountains.
- For professional use only. KEEP OUT OF CHILDRED'S REACH

Section Ⅳ-Physical and Chemical Properties

Melting Point (Softening)	800°	Boiling Point℃	N/A
Specific Gravity (Bare Glass)	2.59	Percent Volatile	N/A
Vapor Pressure (mm/Hg)	N/A	Vapor Density (Air=1)	N/A
Evaporative Rate (Ethy1 Ether=1)	N/A	Solubility in Water	Not Soluble
Appearance and Odor		White/Off-white/tan colored solid with no odor	

PH N/A

Section Ⅴ-Stability and Reactivity

Stability (Conditions to Avoid) Product is stable

Incompatibility (Materials to Avoid) None known

Hazardous Decomposition Products Sizing or binders may decompose in a fire. Primary decomposition products include carbon monoxide, carbon dioxide, other hydrocarbons and water.

Hazardous Polymerization Will not occur.

To the best of our knowledge, the information contained herein is accurate. The information provided is based upon data furnish by Heat Shield. Assume liability whatsoever for the accuracy or completeness of the information contained herein. While believed to be reliable, the information of products is intended for use by skilled persons at their own risk. Final determination of suitability of any material is the sole responsibility of the user. All materials may present unknown hazards and should be used with caution. Although certain hazards are described herein, we cannot guarantee that those are the only hazards that exist.

Material Safety Data Sheet

Section Ⅰ

FIBERGLASS SEWING THREAD COATED WITH PTFE

Section Ⅱ-Composition/Information on Ingredients

Hazardous Ingredients	Weight %	OSHA-PEL	ACGIH-TLV	Other
Fiberglass, continuous filament	≥90	*	$10mg/m^3$ 8-hr TWA	$3X10$ fibers/m^3 10-hr TWA(NIOSH)
PTFE	≥10			
Non-hazardous Ingredients				

*OSHA has not established a specific PEL for fibrous glass. It is considered to be a "particulate not otherwise regulated" (PNOR) and is covered under the OSHA nuisance dust PEL's of 5 mg/m^3 for the respirable dust fraction and 15 mg/m^3 for the total dust fraction for an 8-hr TWA(Time Weighted Average).

Section Ⅲ-Hazardous Identification

Primary Routes of Exposure Inhalation and skin contact

Health Hazards (acute & chronic effects and symptoms of overexposure)

Acute

Inhalation-Inhalation of dusts and fibers may result in irritation of the upper respiratory tract (mouth, nose and throat.)

Skin Contact-Skin contact with fibers and dust may produce temporary mechanical irritations.

Ingestion- Temporary mechanical irritations of the digestive tract. Observe individual. If symptoms develop, consult a physician.

Chronic

See carcinogenicity section below. There is no known health effects associated with chronic exposure to this product.

Carcinogenicity

Hazardous Ingredients	ACGIH	IARC	NTP	OSHA
Fiberglass continuous filament	No	No	No	No

*IARC-In June 1987 the International Agency for research on Cancer(IARC)categorized fibrous continuous filaments as not classifiable with respect to human carcinogenicity(Group 3). The evidence from human as well as animal studies was evaluated by IARC as insufficient to classify fiberglass continuous filaments as a possible, probable, of confirmed cancer causing material.

Medical Conditions Aggravated by Exposure Persons with a history of chronic respiratory or skin conditions that are aggravated by mechanical irritants may be at increased risk for worsening their condition from exposure during use of this product.

Section Ⅳ-First Aid Measures

Inhalation Move individual to fresh air. Seek medical attention if irritation persists.

Skin Contact Wash with mild soap and running water. Use a washcloth to help remove fibers. To avoid further irritation, do not rub or scratch irritated areas. Rubbing or scratching may force fibers into the skin. Seek medical attention if irritation persists.

Eye Contact Flush eyes with flowing water for at less 15 minutes. Seek medical attention if irritation persists.

Ingestion N/A

Section Ⅴ-Fire Aid Measures

Flash Point (°F) N/A

Auto Ignition Temperature(°F) N/A

Flammability Limits (%) LEL: N/A **UEL**: N/A

Extinguishing Media Water, foam, carbon dioxide, dry chemical

Special Fire Fighting Instructions In sustained fire, self-contained breathing apparatus should be worn.

Unusual Fire and Explosion Hazards None known

Section Ⅵ-Accidental Release Measures

Action To Take For Spills For sold product no applicable. For dusts and fibers generated during fabrication, vacuum up and containerize.

Section Ⅶ-Handling, Storage and Disposal

Ventilation General dilution ventilation and /or local exhaust ventilation should be provided, as necessary, to maintain exposures below PEL' S or TLV' S. ADEQUATE VENTILATION MUS BE PROVIDED AT ELEVATED TEMPERATURE.

Respiratory Protection A properly fitted NIOSH/MHSA approved dust respirator such as 3M model 8710 or model 9900(In high humidity environment)or equivalent should be used when: high dust levels are encountered; the level of glass fibers in the air exceeds the OSHA permissible exposure limits; or if irritation occurs. Use respiratory protection in accordance with your company' s respiratory protection program and OSHA regulations under 29 CFR 1910.134.

Eye Protection Safety glasses, goggles or face shields should be worn whenever fiberglass materials are handled.

Work/Hygienic Practices Handle in accordance with good industrial hygiene and safety practices.

- Avoid unnecessary exposure to dusts and fibers.
- Remove fibers from skin after exposure.
- Be careful not to bur or scratch irritated areas. Rubbing or scratching may force the fibers into the skin. The fibers should be washed off. Use of barrier creams can, in some instances, be helpful.
- Use vacuum equipment to remove fibers and dusts from clothing. COMPRESSED AIR SHOULD NEVER BE USED. Always wash work cloths separately and wipe out the washer/sink in order to prevent loose glass fibers from getting in other clothing.
- Keep work area clean of any dust and fibers. Avoid sweeping or using compressed air as these techniques re-suspend dusts and fibers into air.
- Have access to safety showers and eye wash fountains.

- For professional use only. KEEP OUT OF CHILDRED' S REACH

HEAT SHIELD

Section Ⅳ-Physical and Chemical Properties

Melting Point(Softening)	800°	**Boiling Point℃**	N/A
Specific Gravity (Bare Glass)	2.59	**Percent Volatile**	N/A
Vapor Pressure (mm/Hg)	N/A	**Vapor Density (Air=1)**	N/A
Evaporative Rate (Ethyl Ether=1)	N/A	**Solubility in Water**	Not Soluble

Appearance and Odor White/Off-white/tan colored solid with no odor

PH N/A

Section Ⅴ-Stability and Reactivity

Stability (Conditions to Avoid) Product is stable

Incompatibility (Materials to Avoid) None known

Hazardous Decomposition Products Sizing or binders may decompose in a fire. Primary decomposition products include carbon monoxide, carbon dioxide, other hydrocarbons and water.

Hazardous Polymerization Will not occur.

To the best of our knowledge, the information contained herein is accurate. The information provided is based upon data furnish by us. Assume liability whatsoever for the accuracy or completeness of the information contained herein. While believed to be reliable, the information of products is intended for use by skilled persons at their own risk. Final determination of suitability of any material is the sole responsibility of the user. All materials may present unknown hazards and should be used with caution. Although certain hazards are described herein, we cannot guarantee that those are the only hazards that exist.

Material Safety Data Sheets

Section I : Product Revision Date: June 01, 2020

Product Name : **Stainless Steel 304 / 304L / 316 / 316L Fasteners**

Product Description : **Stainless Steel Cord End Clip, Eyelet & Ring, Lacing Anchor, Tie Wire etc.**

This MSDS covers all grades listed above that have additional letter and/or number suffixes or prefixes.

SECTION II – Hazardous Ingredients Information:

Stainless steel products covered by this MSDS are shipped as non-reactive, non-flammable, non-explosive, and essentially non-hazardous materials. The information in this section covers the chemicals from which this product is manufactured. The fumes which may be produced as a by-product during welding, brazing, thermal cutting, or similar processes are discussed in Section V.

HAZARDOUS COMPONENTS[1]:

Specific Chemical Identity (Common Name)	CAS Number	SARA[2]	OSHA PEL[3] (mg/m^3)	ACGIH TLV[4] (mg/m^3)	NOTES	PERCENT[5] By Weight
Chromium	7440-47-3	*		0.01	Cr(VI)	10.5-18.0
			1	0.5	Metal	
Iron	7439-89-6		10	5	(Fe_2O_3)	> 50
Manganese	7439-96-5	*	5[6]	0.2	Fume	≤ or = 1
			5[6]	0.2	Dust	
Nickel	7440-02-0	*	1	1.5	Metal	< or 5
				0.1	Soluble	
				0.2	Insoluble	
Silicon	7440-21-3		15	10	Total dust	≤ or = 1
			5	NA	Respirable	

Notes:

1. The term "Hazardous" should be interpreted as defined and required in the OSHA Hazard Communication Standard (29 CFR 1910.1200) and does not necessarily imply the existence of any hazard. Any components at concentrations equal to or greater than 0.1 percent are listed in this section, according to OSHA 29 CFR 1910.1200.

2. An asterisk (*) after the Chemical Abstract Service (CAS) Number indicates a toxic chemical subject to the reporting requirements of Section 313 of the Emergency Planning and Community Right-To-Know Act of 1986 (SARA) and 40 CFR Part 372.

3. These permissible exposure levels (PELs) are based on OSHA's rulemaking (29 CFR 1910 Subpart Z) adopted on May 29, 197l, and are the current regulatory limits.

4. These values are based on the American Conference of Governmental Industrial Hygienists (ACGIH) 2002 TLVs.

5. Approximate percent by weight values.

6. Ceiling Limit.

HEAT SHIELD

SECTION III - Physical/Chemical Characteristics:

BOILING POINT:	NA	**VAPOR DENSITY (Air = 1):**	NA
MELTING POINT:	NA	**VAPOR PRESSURE (mm Hg):**	NA
SPECIFIC GRAVITY:	NA	**EVAPORATION RATE**	
pH:	NA	**(Butyl Acetate = 1):**	NA
APPEARANCE AND ODOR:	Solid, metallic gray stainless steel; odorless.	**SOLUBILITY IN WATER**	NA

SECTION IV – Fire and Explosion Hazard Data:

FLASH POINT (Method Used):	NA
FLAMMABLE LIMITS:	**LEL:** NA **UEL:** NA
EXTINGUISHING MEDIA:	No special media required.
SPECIAL FIRE FIGHTING PROCEDURES:	None.
UNUSUAL FIRE AND EXPLOSION HAZARDS:	This product is not flammable, combustible or explosive.

SECTION V – Reactivity Data:

STABILITY: **Stable: X Unstable:____**
Conditions to Avoid: None known.

INCOMPATIBILITY (Materials to Avoid): None known.

HAZARDOUS POLYMERIZATION: **May occur:___Will not occur: X__**
Conditions to Avoid: None known.

HAZARDOUS DECOMPOSITION OR BYPRODUCTS: Various fumes and gases may be produced when stainless steel is subjected to welding, brazing, thermal cutting, and similar processes. Such fumes and gases cannot be simply classified. The composition and quantity of both are dependent upon the composition of the base metal and the process, procedures, and consumables being used. Other conditions which also influence the composition and quantity of the fumes and gases to which workers may be exposed include; coatings on the base metal (such as paint, plating, galvanizing, and phosphate coatings), the number of workers performing welding, brazing, thermal cutting, or other related operations, the volume of the work area, the quantity of consumables used, the design and amount of ventilation delivered, the position of the worker's head with respect to the fume plume, and the presence of contaminants in the atmosphere (such as chlorinated hydrocarbon vapors from solvent, cleaning, or painting activities) which may decompose by the arc into toxic gases such as phosgene.

Decomposition products from welding, brazing, thermal cutting operations will include those originating from the volatilization, reaction, or oxidation of ingredients in welding rods, fluxes, and fillers, plus those from the base metal and coatings, etc. Possible decomposition products that may be generated during welding, brazing or thermal cutting include complex oxides of the ingredients listed in Section II. Fumes generated during welding, brazing, or thermal cutting may contain: chromium compounds, including hexavalent chromium (Cr VI); nickel; manganese; iron; and silicon compounds.

The employer is required by OSHA to limit the worker's level of exposure to chemicals for which OSHA has established a PEL in 29 CFR 1910 Subpart Z. The only way to determine a worker's exposure to welding, brazing or thermal cutting decomposition products is by sampling and analyses using accepted industrial hygiene techniques. The composition and quantity of the fumes and gases to which a worker is exposed can be established from an air samples(s) obtained from inside the welder's helmet, if worn, or in the worker's breathing zone. Review ANSI/AWS F1.l and F1.3 standards for further information on air sampling for welding decomposition products.

HEAT SHIELD

SECTION VI – Health Hazard Data:

ROUTE(S) OF ENTRY: Inhalation: Unlikely **Skin/Eye Contact:** Yes **Skin Absorption:** Unlikely **Ingestion:** Unlikely

As distributed, stainless steel does not present an inhalation, absorption, or ingestion hazard. However, operations such as welding, brazing, thermal cutting, burning, sawing or grinding may result in the following health effects:

HEALTH HAZARDS (Acute and Chronic):

Acute (Short-Term Effects): Short-term exposures to the fumes generated by welding, brazing, or thermal cutting on stainless steel may result in dizziness; nausea; and irritation of the eyes, skin, lungs, nose and throat. Metal fume fever, a flu-like illness lasting about 24 hours with chills, aches, cough, and fever can be caused by overexposure to metal fumes, including iron, chromium, and manganese.

Metal dust particles may cause eye, skin and/or respiratory system irritation. Acute asthma attacks may be experienced by asthmatics when metal dust or fume is inhaled. Skin contact with chromium and nickel-containing dust may cause allergic contact dermatitis on sensitized individuals.

Chronic (Long-Term Effects): Long-term exposures to metal dusts or fumes generated by welding, brazing, thermal cutting, burning, sawing, grinding or otherwise processing this product may lead to the accumulation of particulate in the lungs and may result in lung fibrosis, or pneumoconiosis. The severity of these conditions depends on the level and duration of the exposure. Overexposure to iron oxide can cause siderosis (deposits of iron in the lungs) which may affect pulmonary function. Select chromium compounds and chromium VI, which may be formed during welding, brazing or thermal cutting on this material, are considered carcinogenic to humans. Nickel metal and compounds that may be formed during welding, brazing, or thermal cutting are known to be carcinogenic. Chromium and nickel compounds can cause lung cancers. Nickel may also cause nasal cancers. Overexposure to chromium compounds may cause respiratory irritation with symptoms resembling asthma, and ulceration and perforation of the nasal septum. Long-term overexposure to nickel fumes can cause allergic asthma, pneumonitis, pulmonary fibrosis and edema. Skin contact with chromium and nickel may cause sensitization. Overexposure to manganese can result in central nervous system effects referred to as manganism, including symptoms of muscular weakness and tremors similar to Parkinson's disease.

CARCINOGENICITY: NTP: Yes* **IARC:** Yes***OSHA REGULATED:** Yes* **CAL. PROP. 65:** Yes**

*Nickel metal and select nickel and chromium compounds are listed in the National Toxicology Program (NTP) Annual Report on Carcinogens and in the International Agency for Research on Cancer (IARC) Monographs and are also considered carcinogens by OSHA.

****WARNING:** This product contains or produces chemicals known to the State of California and other states, if applicable, to cause cancer.

SIGNS AND SYMPTOMS OF EXPOSURE: Inhalation of fumes may cause irritation of the lungs, damage to the lungs and asthma-like symptoms. Overexposure to metals in fumes can cause metal-fume fever which is a flu-like sickness including chills, fever, head and muscle ache, tightness of chest, dryness of nose and mouth, nausea and vomiting. Symptoms occur within several hours of exposure lasting 6-24 hours. Extreme overexposure can cause death. Chromium dust can burn eyes and form ulcers on skin. Absorption through the skin can affect kidney and liver function. Manganese can produce behavioral changes and changes in hand writing. Exposure to metal dust or fume may cause irritation to eyes, skin, and/or respiratory system. Allergic contact dermatitis may occur in individuals who are sensitized to chromium or nickel.

MEDICAL CONDITIONS GENERALLY AGGRAVATED BY EXPOSURE: Exposure to metals dust and fumes may aggravate pre-existing respiratory or allergic conditions in some workers. Some studies have shown a higher level of lung related problems among order welders who smoked than those who did not smoke.

EMERGENCY AND FIRST AIR PROCEDURES: Remove from exposure area and call for medical aid. Employ first aid techniques recommended by the American Red Cross.

Eyes: In case of irritation from particulate, immediately flush with plenty of water for 15 minutes and call for medical assistance.

Skin: In case of skin contact causing irritation, wash thoroughly with plenty of soap and water.

Inhalation: If breathing is difficult, remove to fresh air and give oxygen. If breathing has stopped, provide artificial respiration.

Ingestion: Accidental ingestion unlikely. If ingested, call for medical assistance.

SECTION VII – Control Measures:

As distributed, control measures are not required while handling stainless steel.

Appropriate control measures should be employed when welding, brazing, thermal cutting, burning, sawing or grinding stainless steel. For welding and related processes, read and understand the MSDSs, manufacturers instruction, and precautionary labels for welding consumables. See American National Standard Z49.1, Safety in Welding and Cutting, published by the American Welding Society, 550 N.W. LeJeune Road, Miami, Florida. 33126 and OSHA Publication 2206 (29 CFR 1910), U.S. Government Printing Office, Washington, D.C. 20402, for more details on exposure controls.

VENTILATION: Use local exhaust and dilution ventilation to control dust and/or fumes during welding, brazing, thermal cutting, burning sawing or grinding on stainless steel pipe and tube.

RESPIRATORY PROTECTION (Specific Type): Use a NIOSH approved respirator for dust and fumes or an air supplied respirator where local exhaust or general dilution ventilation does not keep exposures below the PEL or TLV for air contaminants.

PROTECTIVE GLOVES: Wear protective gloves while handling stainless steel to prevent cuts and skin abrasions, and to reduce the risk of sensitization from skin contact. For activities such as welding, brazing and thermal cutting, appropriate hand protection should be worn.

EYE PROTECTION: ANSI Z87.1 approved safety glasses with side shields or goggles should be worn where metal dust or fume is present. Use appropriate eye protection, including welding helmets and/or face shields with protective filter lenses when welding, brazing or thermal cutting. Select welding lens shades from the American Welding Society (AWS) publication F2.2.

PROTECTIVE CLOTHING: Protective clothing is generally not required when handling stainless steel. Remove and launder work clothes soiled with metal dust to avoid spreading contamination outside the work area. Wear appropriate head and body protection during welding, brazing and thermal cutting on stainless steel. Refer to ANSI Z49.1 for more information.

OTHER PROTECTIVE EQUIPMENT: Protective clothing such as uniforms, disposable coveralls, safety shoes, etc. may be required during metal handling operations as appropriate.

WORK/HYGIENIC PRACTICES: Do not eat, smoke, or drink in areas where metal dust or fume is present. Utilize good personal hygiene including washing hands and face prior to eating or drinking.

SECTION VIII – Precautions for Safe Handling and Use:

STEPS TO BE TAKEN IN CASE MATERIAL IS RELEASED OR SPILLED: In solid form this material poses no special clean-up problems. If this material is in powder or dust form, clean up should be conducted with a vacuum system utilizing a high efficiency particulate air filtration system. Caution should be taken to minimize airborne generation of powder or dust and avoid contamination of air and water. Place material in a suitable waste container and properly label all containers.

WASTE DISPOSAL METHOD: Dispose of product, residue, liners, containers and waste material in accordance with local, state, and federal regulations.

OTHER PRECAUTIONS: Use good personal hygiene and safe work practices. Keep out of reach of children. Always read and follow directions on product label and other product information.

ENVIRONMENTAL HAZARDS: In solid form this material poses no special environmental problems. Metal powders or dusts may have significant impact on air and water quality. Airborne emissions, spills and releases to the environment (discharge to streams, sewer systems, ground water, surface soil, etc.) should be controlled immediately. If such potential for a spill or release exists, it is advisable to develop an emergency spill response plan.

SECTION IX – California Proposition 65:

END OF MSDS.